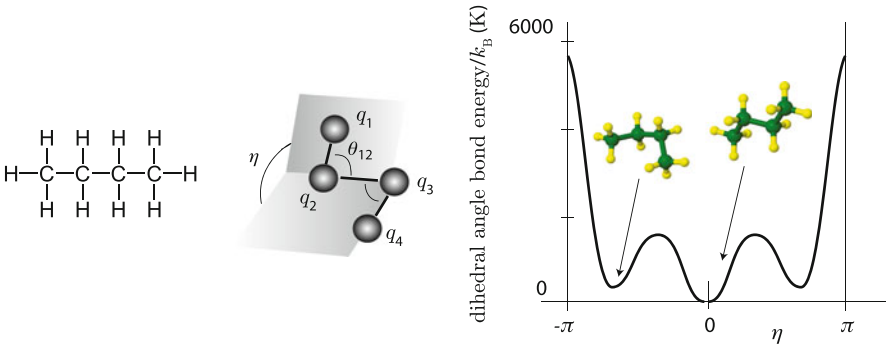


Fig. 1.10 *Left*: structural diagram of butane. *Center*: coarse-grained model of butane using CH_3 and CH_2 groups. *Right*: torsion (or dihedral) potential energy showing locations of local minimum corresponding to trans (global minimum) and gauche (secondary local minimum) states

secondary local minimum for the *gauche* (compact) state (see Fig. 1.10). The same approach can be used to treat many types of large flexible chain molecules.

When a liquid or gaseous alkane system is simulated, the potential energy will include all those intra-molecular terms as well as non-bonded interactions between the atoms of different butane molecules, which might be modelled by Lennard-Jones or, more commonly in recent treatments of alkanes, Buckingham potentials. The potential energy function for an alkane then consists of sums over all the potential energy terms. There are many choices for the detailed specification of parameters. See [340] for a well described model for n-alkane chains.

Example 1.2 (Gay-Berne Potential) In simulations of liquid crystals and other systems with a coarse-grained molecular dynamics model, it is common to replace groups of atoms by ellipsoidal rigid bodies. To correctly model the short ranged interactions of such objects, one needs to introduce an anisotropic intermolecular pair potential to model the short-ranged interactions Fig. 1.11a. A common choice for this purpose is the *Gay-Berne* potential [143] (and various generalizations) which takes into consideration the separation between ellipsoids as well as their orientations relative to the vector linking their centers of mass. The situation is diagrammed in Fig. 1.11b, where the centers of mass of two molecules are $\boldsymbol{q}_1$ and $\boldsymbol{q}_2, \boldsymbol{q}_{12} = \boldsymbol{q}_2 - \boldsymbol{q}_1$ being the separation vector. $\hat{\boldsymbol{u}}_1$ and $\hat{\boldsymbol{u}}_2$ are unit vectors representing the orientations of the molecules. The potential energy may be written as

$$\varphi_{\mathrm{GB}}(\boldsymbol{q}_1, \boldsymbol{q}_2, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2) = 4\epsilon_{\mathrm{GB}} \left[\left(\frac{\sigma_0}{\Delta} \right)^{12} - \left(\frac{\sigma_0}{\Delta} \right)^{6} \right],$$

$$\epsilon_{\mathrm{GB}}(\hat{\boldsymbol{r}}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2) = \epsilon_1(\hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2) \epsilon_2(\hat{\boldsymbol{r}}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2),$$

with

$$\Delta(\boldsymbol{r}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2) = \|\boldsymbol{r}\| - \sigma_0 / \sqrt{W(\hat{\boldsymbol{r}}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2, \chi)},$$

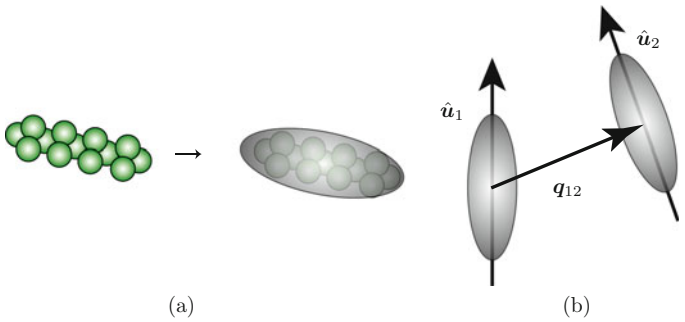


Fig. 1.11 (**a**) A typical coarse-graining method replaces groups of atoms by ellipsoidal rigid bodies. (**b**) The Gay-Berne potential models the anisotropic interaction between nonspherical coarse-grained molecules; potentials are described by the orientation of the molecules with respect to a fixed frame and the separation between their centers of mass

$$\epsilon_1(\hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2) = \epsilon_0 \left[1 - \chi^2 (\hat{\boldsymbol{u}}_1 \cdot \hat{\boldsymbol{u}}_2)^2\right]^{-1/2},$$
$$\epsilon_2(\hat{\boldsymbol{r}}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2) = W(\hat{\boldsymbol{r}}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2, \chi'),$$

and

$$W(\hat{\boldsymbol{r}}, \hat{\boldsymbol{u}}_1, \hat{\boldsymbol{u}}_2, \chi) \stackrel{\text{def}}{=} 1 - \frac{\chi}{2} \left[\frac{(\hat{\boldsymbol{r}} \cdot (\hat{\boldsymbol{u}}_1 + \hat{\boldsymbol{u}}_2))^2}{1 + \chi \hat{\boldsymbol{u}}_1 \cdot \hat{\boldsymbol{u}}_2} + \frac{(\hat{\boldsymbol{r}} \cdot (\hat{\boldsymbol{u}}_1 - \hat{\boldsymbol{u}}_2))^2}{1 - \chi \hat{\boldsymbol{u}}_1 \cdot \hat{\boldsymbol{u}}_2} \right].$$

Finally

$$\chi = \frac{[\sigma_e/\sigma_s]^2 - 1}{[\sigma_e/\sigma_s]^2 + 1}, \qquad \chi' = \frac{1 - [\epsilon_e/\epsilon_s]^{1/\mu}}{1 + [\epsilon_e/\epsilon_s]^{1/\mu}}.$$

In the original paper, the parameters were taken to be

$$\sigma_e/\sigma_s = 3, \qquad \epsilon_e/\epsilon_s = 0.2, \qquad \mu = 2,$$

yielding $\chi = 4/5$, $\chi' = 0.38$.

The Gay-Berne potential is similar to a Lennard-Jones interaction, but the well depth and its location are adjusted according the orientations of the molecules relative to the vector $\boldsymbol{q}_{12}$. Figure 1.12 shows the appearance of the Gay-Berne potential energy for two different alignments of the rigid bodies.

Let us emphasize that the design and parameterization of force fields for complicated molecular systems is a difficult task as the properties of the system may only be realized via extensive simulation. In particular, where molecular models are used to make predictions in thermodynamic regimes that are far from available experimental data, the results can occasionally be unpredictable. For example, a NAMD [302] case study simulating the melting of ice demonstrates the care needed